

Examples Sheet 1

The starred questions are intended as extras: do them if you have time, but not at the expense of unstarred questions on later sheets.

1. Show, from first principles, that, for non-negative integer n , $d^n x^n / dx^n = nx^{n-1}$.
2. Let $f(x) = u(x)v(x)$. Use the definition of the derivative of a function to show that

$$\frac{df}{dx} = u \frac{dv}{dx} + \frac{du}{dx} v.$$

3. Calculate:

- (i) $d^n(xe^x)/dx^n$ using (a) the Leibniz rule and (b) application of the product rule;
- (ii) $d^3(\ln x)^2/dx^3$.

4. (i) Write down or determine the Taylor series for $f(x) = e^{ax}$ about $x = 1$.
- (ii) Write down or determine the Taylor series for $\ln(1+x)$ about $x = 0$. Then show that

$$\lim_{k \rightarrow \infty} k \ln \left(1 + \frac{x}{k} \right) = x$$

and deduce that

$$\lim_{k \rightarrow \infty} \left(1 + \frac{x}{k} \right)^k = e^x.$$

5. Determine by any method the first three non-zero terms of the Taylor expansions about $x = 0$ of:

- (i) $(x^2 + a)^{-3/2}$;
- (ii) $\ln(\cos x)$;
- (iii) $\exp[-(x-a)^{-2}]$, where a is a constant.

6. By considering the area under the curves $y = \ln x$ and $y = \ln(x - 1)$, show that

$$N \ln N - N < \ln(N!) < (N + 1) \ln(N + 1) - N.$$

Hence show that

$$\frac{N!}{N + 1} < \left(\frac{N + 1}{e} \right)^N.$$

7. Show that

$$y(x) = \int_x^\infty e^{-t^2} dt$$

satisfies the differential equation $y'' + 2xy' = 0$.

*8. Let J_n be the indefinite integral

$$J_n = \int \frac{x^{-n}}{(ax^2 + 2bx + c)^{\frac{1}{2}}} dx.$$

By integrating $\int x^{-(n+1)}(ax^2 + 2bx + c)^{\frac{1}{2}} dx$ by parts, show that, for $n \neq 0$,

$$ncJ_{n+1} + (2n - 1)bJ_n + (n - 1)aJ_{n-1} = -x^{-n}(ax^2 + 2bx + c)^{\frac{1}{2}}.$$

Hence evaluate

$$\int_1^2 \frac{1}{x^{5/2}(x + 2)^{\frac{1}{2}}} dx.$$

*9. In a large population, the proportion with income between x and $x + dx$ is $f(x)dx$. Express the mean (average) income μ as an integral, assuming that any positive income is possible.

Let $p = F(x)$ be the proportion of the population with income less than x , and $G(x)$ be the mean (average) income earned by people with income less than x . Further, let $\theta(p)$ be the proportion of the total income that is earned by people with income less than x as a function of the proportion p of the population that has income less than x . Express $F(x)$ and $G(x)$ as integrals and thence derive an expression for $\theta(p)$, showing that

$$\theta(0) = 0, \quad \theta(1) = 1$$

and

$$\theta'(p) = \frac{F^{-1}(p)}{\mu}, \quad \theta''(p) = \frac{1}{\mu f(F^{-1}(p))} > 0.$$

(You may assume that $F(x)$ is a strictly increasing function of x .)

Sketch the graph of a function $\theta(p)$ with these properties. Deduce that, if there is any variation in income, the bottom (when ordered in terms of income) proportion p of the population receive less than p of the total income, for all positive values of p . Just how much less is quantified by the (in)famous “Gini index” beloved of economists, which is twice the area between the curve $\theta(p)$ and the diagonal line connecting $(0, 0)$ and $(1, 1)$.

A particular population's income is described by the exponential distribution:

$$f(x) = \lambda e^{-\lambda x} \quad \text{for } x > 0.$$

For some constant $\lambda > 0$, using your expression for $\theta(p)$, compute the Gini index in this case.

[Food for thought: To what extent does the Gini index capture the degree of income inequality for this distribution?]

10. For $f(x, y) = \exp(-xy)$, find $(\partial f / \partial x)|_y$ and $(\partial f / \partial y)|_x$. Check that

$$\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial^2 f}{\partial y \partial x}.$$

Now express x and y in polar coordinates, $x = r \cos \theta$ and $y = r \sin \theta$. Find $(\partial f / \partial r)|_\theta$ and $(\partial f / \partial \theta)|_r$ using:

- (a) the chain rule; and
- (b) by first expressing f in terms of r and θ .

Check that the two methods give the same results.

11. If $xyz + x^3 + y^4 + z^5 = 0$ (an implicit equation for any of the variables x, y, z in terms of the other two), find

$$\left. \frac{\partial x}{\partial y} \right|_z, \quad \left. \frac{\partial y}{\partial z} \right|_x \quad \text{and} \quad \left. \frac{\partial z}{\partial x} \right|_y$$

and show that their product is -1 .

Does this result hold for an arbitrary relation $f(x, y, z) = 0$?

What about $f(x_1, x_2, \dots, x_n) = 0$?

12. In thermodynamics, the pressure of a system, p , can be considered as a function of the variables V (volume) and T (temperature) or as a function of V and S (entropy).

- (i) By expressing $p(V, S)$ in the form $p(V, S(V, T))$ evaluate

$$\left. \frac{\partial p}{\partial V} \right|_T - \left. \frac{\partial p}{\partial V} \right|_S \quad \text{in terms of} \quad \left. \frac{\partial S}{\partial V} \right|_T \quad \text{and} \quad \left. \frac{\partial S}{\partial p} \right|_V.$$

- (ii) Hence, using $TdS = dU + pdV$ (conservation of energy with U the internal energy), show that

$$\left. \frac{\partial \ln p}{\partial \ln V} \right|_T - \left. \frac{\partial \ln p}{\partial \ln V} \right|_S = \left. \frac{\partial(pV)}{\partial T} \right|_V \left[\frac{p^{-1} (\partial U / \partial V)|_T + 1}{(\partial U / \partial T)|_V} \right].$$

[Hint: note that $(\partial \ln p / \partial \ln V)|_T = (V/p) (\partial p / \partial V)|_T$.]

13. By differentiating I with respect to λ , show that

$$I(\lambda, \alpha) = \int_0^\infty \frac{\sin \lambda x}{x} e^{-\alpha x} dx = \tan^{-1} \left(\frac{\lambda}{\alpha} \right) + c(\alpha).$$

Show that $c(\alpha)$ is constant (independent of α) and hence, by considering the limits $\alpha \rightarrow \infty$ and $\alpha \rightarrow 0$, show that, if $\lambda > 0$,

$$\int_0^\infty \frac{\sin \lambda x}{x} dx = \frac{\pi}{2}.$$

What is the value of the integral when $\lambda < 0$?

14. Let

$$f(x) = \left(\int_0^x e^{-t^2} dt \right)^2 \quad \text{and} \quad g(x) = \int_0^1 \frac{e^{-x^2(t^2+1)}}{(1+t^2)} dt.$$

Show that

$$f'(x) + g'(x) = 0.$$

Deduce that

$$f(x) + g(x) = \pi/4,$$

and hence that

$$\int_0^\infty e^{-t^2} dt = \frac{\sqrt{\pi}}{2}.$$

Comments and corrections may be sent by email to c.e.thomas@damtp.cam.ac.uk